# Evaluating the Generalizability of Heart Disease Prediction Models Across Heterogeneous Populations

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**Abstract**

Machine learning (ML) models have shown strong performance in predicting cardiovascular disease. However, concerns remain about whether these models generalize well across different patient populations. Many studies report high accuracy using internal validation, but methodological guidelines emphasize that this is not sufficient for real-world clinical use. Previous research shows that model performance often declines when applied to data from different healthcare systems. In addition, distribution shift—where new data differs from training data—can reduce model reliability and calibration.

This study evaluates the generalizability of heart disease prediction models using the UCI Heart Disease dataset, which includes four cohorts: Cleveland, Hungary, Switzerland, and VA Long Beach. Models are trained on a merged dataset (Cleveland and Hungary) and evaluated on external cohorts (Switzerland and VA) to assess performance changes across populations.

The findings show a consistent decrease in performance under external validation, highlighting the impact of distribution shift on model generalization. The results demonstrate a clear degradation in performance under cross-cohort conditions and emphasize the importance of external validation for developing reliable medical prediction models.

**Keywords:** Machine learning, Cardiovascular disease, External validation, Distribution shift

# 1 Introduction

Cardiovascular disease (CVD) continues to place a significant burden on global health, contributing to high rates of morbidity and mortality. To address this, predictive models are increasingly used to support early diagnosis and improve risk assessment. In recent years, machine learning (ML) techniques have been applied to structured clinical data to enhance cardiovascular prediction tasks [5]. Research in the Journal of the American College of Cardiology further highlights the expanding role of ML models in cardiovascular diagnostics and risk evaluation [10].

Several studies report that supervised learning algorithms, including ensemble methods, can achieve high classification accuracy on structured cardiovascular datasets [13]. However, many of these studies depend on internal validation approaches, such as random train–test splits within the same dataset, which may not reflect real-world performance.

Methodological guidelines published in BMJ emphasize that this type of evaluation may overestimate real-world performance and recommend external validation across independent populations [1]. Empirical research reported in JAMA Internal Medicine shows that predictive performance often decreases when models are applied to data from different healthcare systems [6]. Similarly, studies in Annals of Internal Medicine indicate that insufficient external validation can negatively affect model calibration and discrimination [14].

The UCI Heart Disease dataset provides a useful opportunity to investigate this problem. It combines clinical data from four independent cohorts located in Cleveland, Hungary, Switzerland, and VA Long Beach [19]. This multi-cohort structure enables direct evaluation of model performance across different populations in a controlled experimental setting, allowing researchers to assess how well prediction models generalize across heterogeneous populations.

This study aims to evaluate how machine learning models perform under internal and external validation and to compare their generalization ability across heterogeneous patient populations.

# 2 Literature Review

## 2.1 Machine Learning in Cardiovascular Risk Prediction

Machine learning (ML) methods have shown promising results in predicting cardiovascular disease and related outcomes. Research published in JAMA Cardiology indicates that ML-based risk prediction models can sometimes perform better than traditional clinical scoring systems when predicting cardiovascular events [5]. In addition, a review in the Journal of the American College of Cardiology highlights the increasing use of advanced data-driven methods to improve cardiovascular risk assessment across different patient groups [10].

Other studies have also reported strong predictive results using ML algorithms. For example, research in Scientific Reports shows that classifiers such as random forests and gradient boosting can achieve high accuracy when applied to structured cardiovascular datasets [13]. However, many of these studies evaluate their models

using internal validation methods, such as dividing the same dataset into training and testing subsets. As a result, the reported performance may not fully represent how the model performs in different populations.

## 2.2 External Validation and Model Transportability

External validation is widely considered necessary for determining whether predictive models are reliable in real clinical settings. Guidelines published in BMJ emphasize that prediction models should be tested on independent cohorts before they are used in practice [1].

Empirical evidence supports this recommendation. A large study reported in JAMA Network Open found that predictive models can experience noticeable performance drops when applied to data from different healthcare systems [7]. Similar results were reported in JAMA Internal Medicine, where diagnostic models showed lower predictive accuracy when tested outside their original development environment [6].

Research published in BMC Medical Research Methodology highlights several statistical challenges when validating prediction models across multiple cohorts, such as differences in patient characteristics and changes in model calibration [2]. Additional evidence from Annals of Internal Medicine highlights the importance of assessing whether models can maintain reliable performance when applied to new patient populations [14].

Overall, these studies show that high accuracy during internal validation does not necessarily mean that a model will perform well in other populations.

## 2.3 Distribution Shift in Clinical AI

Distribution shift occurs when the characteristics of the data used for model training differ from those encountered during deployment. In healthcare applications, these differences may arise due to variations in patient demographics, clinical practices, or measurement procedures.

Research published in npj Digital Medicine demonstrates that even small changes in data distribution can negatively affect model performance and calibration [11]. More recent work also proposes monitoring strategies to detect and manage model drift after AI systems are deployed in clinical environments [12].

Studies reported in the Journal of Medical Internet Research also note that healthcare AI systems are especially vulnerable to domain shift because patient populations often vary across institutions [9]. Similarly, findings from JAMIA Open emphasize the importance of evaluating predictive models across multiple healthcare sites before clinical use [8].

Additional studies indexed in PubMed Central (PMC10513895; PMC8478611) show that prediction accuracy can decrease when models are applied to data from different institutions without recalibration. These results further highlight the importance of conducting cross-cohort validation experiments [15, 16].

### 2.4 Ethical and Clinical Implementation Considerations

In addition to predictive performance, AI systems used in healthcare must also be evaluated for fairness and potential bias. Research in Frontiers in Digital Health discusses ethical concerns related to deploying predictive models without reliable validation across different patient groups [4].

Evidence available through PubMed Central (PMC9281065; PMC12525484) also stresses the need to evaluate model performance across demographic subgroups to ensure fairness in clinical decision-making [17, 18]. Furthermore, research published in European Heart Journal – Digital Health recommends that cardiovascular AI systems should be validated across diverse geographic populations before being integrated into clinical practice [3].

Together, these studies highlight that ensuring model reliability across different populations is both a technical requirement and an ethical responsibility.

### 2.5 The UCI Heart Disease Dataset

For the analysis, this research relies on the UCI Heart Disease dataset, which contains patient information gathered from four separate cohorts: Cleveland, Hungary, Switzerland, and VA Long Beach [19].

Because the dataset includes data from multiple populations, it provides a useful setting for evaluating whether prediction models can perform consistently across different cohorts. While many previous studies have focused only on the Cleveland subset, using all four cohorts allows the study to examine how well models perform when applied to different patient populations.

### 2.6 Evaluation Metrics in Medical Prediction Models

In studies that apply machine learning to medical prediction, several evaluation metrics are used to measure model performance. Metrics such as accuracy, precision, recall, F1-score, and the area under the receiver operating characteristic curve (ROC-AUC) are frequently used to evaluate predictive performance in medical machine learning research [1, 13, 14].

While previous studies report strong predictive performance using internal validation, their reliance on single-cohort datasets limits generalizability. Additionally, although external validation is recommended, many studies do not implement cross-cohort evaluation, highlighting a gap between methodological recommendations and practical application.

## 3 Research Gap

Machine learning models have shown strong performance in predicting cardiovascular disease [5, 10]. However, there are still important limitations in how these models are evaluated. Many studies report results based mainly on internal validation methods, which may not represent how the models perform in real clinical environments [1].

Previous research has shown that model performance often decreases when models are tested on data from different healthcare systems [6, 7]. In addition, distribution

shift, when the characteristics of new data differ from the training data, can affect model calibration and predictive accuracy when applied to diverse patient populations [11, 12]. Although methodological studies highlight the importance of validating models across multiple cohorts [2], many cardiovascular ML studies still use data from only a single cohort.

Even when datasets include multiple cohorts, such as the UCI Heart Disease dataset, many previous studies use only the Cleveland subset. This limits the ability to evaluate whether the models can perform well across different populations. Although some studies discuss implementation and fairness concerns [3, 4], there are still few experiments that evaluate model performance across different populations.

Therefore, there is a clear gap in systematically evaluating how heart disease prediction models generalize across multiple cohorts within the same dataset using both internal and external validation strategies. Addressing this gap is essential for understanding the reliability of machine learning models in real-world clinical settings.

# 4 Research Questions and Hypotheses

## 4.1 Research Questions

Based on the research gap identified in the literature, this study investigates the following research questions:

- RQ1: How does the predictive performance of heart disease models differ when they are trained and tested on the same cohort compared to when they are trained and tested on different cohorts?
- RQ2: How does the performance of machine learning models trained on a merged cohort change when evaluated on external cohorts in terms of accuracy, precision, recall, F1-score, and ROC-AUC?
- RQ3: Do some machine learning algorithms perform more reliably than others when applied to different patient populations?

These research questions are based on previous studies showing that predictive performance often decreases when models are applied to data from different healthcare systems [6, 7]. In addition, prior research has shown that distribution shift can significantly affect the stability and reliability of clinical machine learning models [11].

## 4.2 Hypotheses

Based on findings reported in previous studies, the following hypotheses are proposed:

- H1: Models evaluated using internal validation (training and testing within the same cohort) will show higher predictive performance compared to models evaluated across different cohorts.

This expectation is supported by research showing that external validation often reveals performance decreases that are not observed during model development [1, 6].

- H2: Model performance will decrease when evaluated on external cohorts compared to internal validation, as reflected by lower accuracy, precision, recall, F1-score, and ROC-AUC values.

This hypothesis is supported by studies showing that changes in data distribution can reduce the reliability of clinical prediction models [11, 12].

- H3: Ensemble-based machine learning algorithms will be more robust to differences between populations compared to simpler linear models.

This hypothesis is based on previous studies that report differences in predictive performance between algorithm types in cardiovascular prediction tasks [10, 13].

# 5 Methodology

## 5.1 Dataset

This study used the UCI Heart Disease dataset, which contains clinical data collected from four independent cohorts: Cleveland, Hungary, Switzerland, and VA Long Beach. Each cohort represents a different patient population and includes clinical attributes such as age, blood pressure, and exercise-related measures.

The original dataset contains a target variable representing different levels of heart disease severity. For the purposes of this study, the target variable was converted into a binary classification problem, where patients with any level of heart disease were labeled as positive cases and patients without heart disease were labeled as negative cases. Using multiple cohorts allowed the study to evaluate whether models trained on one population could generalize to other populations.

## 5.2 Data Preprocessing

Several preprocessing steps were performed to ensure data quality and consistency across the cohorts. First, dataset columns were converted to numeric format where applicable to ensure compatibility with machine learning algorithms. Each dataset was examined for duplicate records, and duplicated entries were removed before further analysis.

Next, categorical features were validated to ensure that values fell within predefined valid categories. Values outside the allowed ranges were treated as missing values. Numeric variables were also checked for unrealistic or impossible values based on predefined thresholds. Values outside these ranges were replaced with missing values. Outliers were also examined using the interquartile range (IQR) method.

A binary target variable indicating the presence or absence of heart disease was then created. Rows with missing target values were removed prior to model training. During model training, missing feature values were handled using median imputation for numeric variables and most-frequent imputation for categorical variables within the preprocessing pipeline. In addition, several features with high missingness or lower stability across cohorts were removed prior to modeling, including ca, chol, thal, fbs, trestbps, restecg, and slope. These features were removed due to high missingness and potential inconsistency across cohorts, which could negatively affect

model generalization. After preprocessing, the final dataset consisted of 7 features used for model training.

## 5.3 Machine Learning Models

Logistic Regression was selected as a baseline model due to its simplicity and interpretability in medical applications, while Random Forest was chosen for its ability to capture nonlinear relationships and complex feature interactions. This combination allows for a comparison between simple and complex models in terms of generalization performance.

A preprocessing pipeline was implemented to standardize the modeling process. Numeric features were scaled using StandardScaler, while categorical features were encoded using one-hot encoding. This preprocessing pipeline ensured that the same data transformations were applied consistently during both model training and testing.

## 5.4 Evaluation Metrics

Model performance was evaluated using standard metrics for binary classification, including accuracy, precision, recall, F1-score, and ROC-AUC. Accuracy measures the proportion of correctly classified instances. Precision reflects the proportion of predicted positive cases that are truly positive, while recall measures the model's ability to correctly identify actual positive cases. The F1-score provides a balance between precision and recall. ROC-AUC evaluates the model's ability to distinguish between positive and negative classes across different classification thresholds.

## 5.5 Experimental Design

Two types of validation experiments were conducted: internal validation and external validation.

For internal validation, each cohort dataset was split into training and testing subsets using a 70% training and 30% testing split. Models were trained and evaluated within the same cohort to measure baseline predictive performance.

For external validation, the Cleveland and Hungary cohorts were combined to create a larger training set. Models were then evaluated on the Switzerland and VA cohorts as external test sets. This design was used to assess how well models trained on a merged population generalize to different patient populations.

This experimental design directly addresses RQ1 by comparing model performance under internal and external validation. RQ2 is evaluated through cross-cohort testing using multiple performance metrics, and RQ3 is examined by comparing the generalization performance of Logistic Regression and Random Forest.

## 5.6 Data Visualization

To better understand the dataset, several visualizations were used during the analysis.

First, the class distribution was analyzed to identify any imbalance between patients with and without heart disease. This step is important because class imbalance can affect model performance and evaluation.

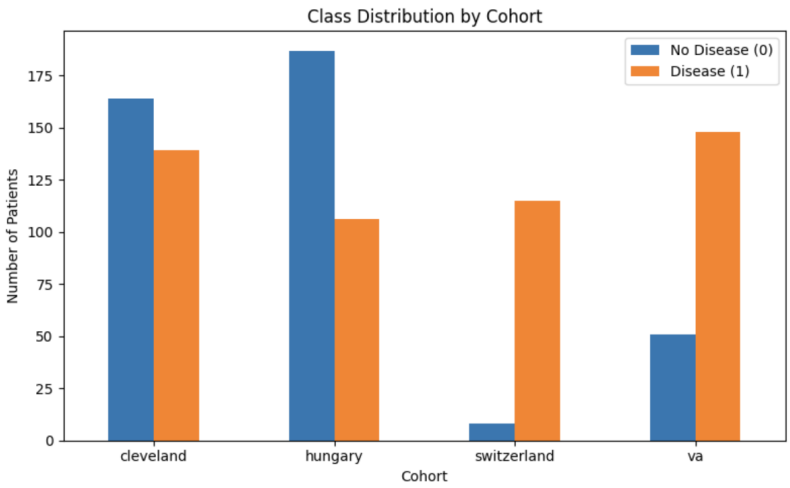


**Fig. 1** Class distribution across cohorts showing the number of patients with and without heart disease in each dataset.

Figure 1 shows the class distribution across the four cohorts. The Switzerland and VA cohorts exhibit a higher proportion of positive cases compared to Cleveland and Hungary, indicating class imbalance. This imbalance may influence model performance, particularly metrics such as recall and F1-score.

Next, histograms and categorical bar plots were used to explore key features such as age, maximum heart rate (thalach), chest pain type (cp), and sex. These visualizations helped identify differences in feature distributions across cohorts, as well as potential skewness and variability in the data.

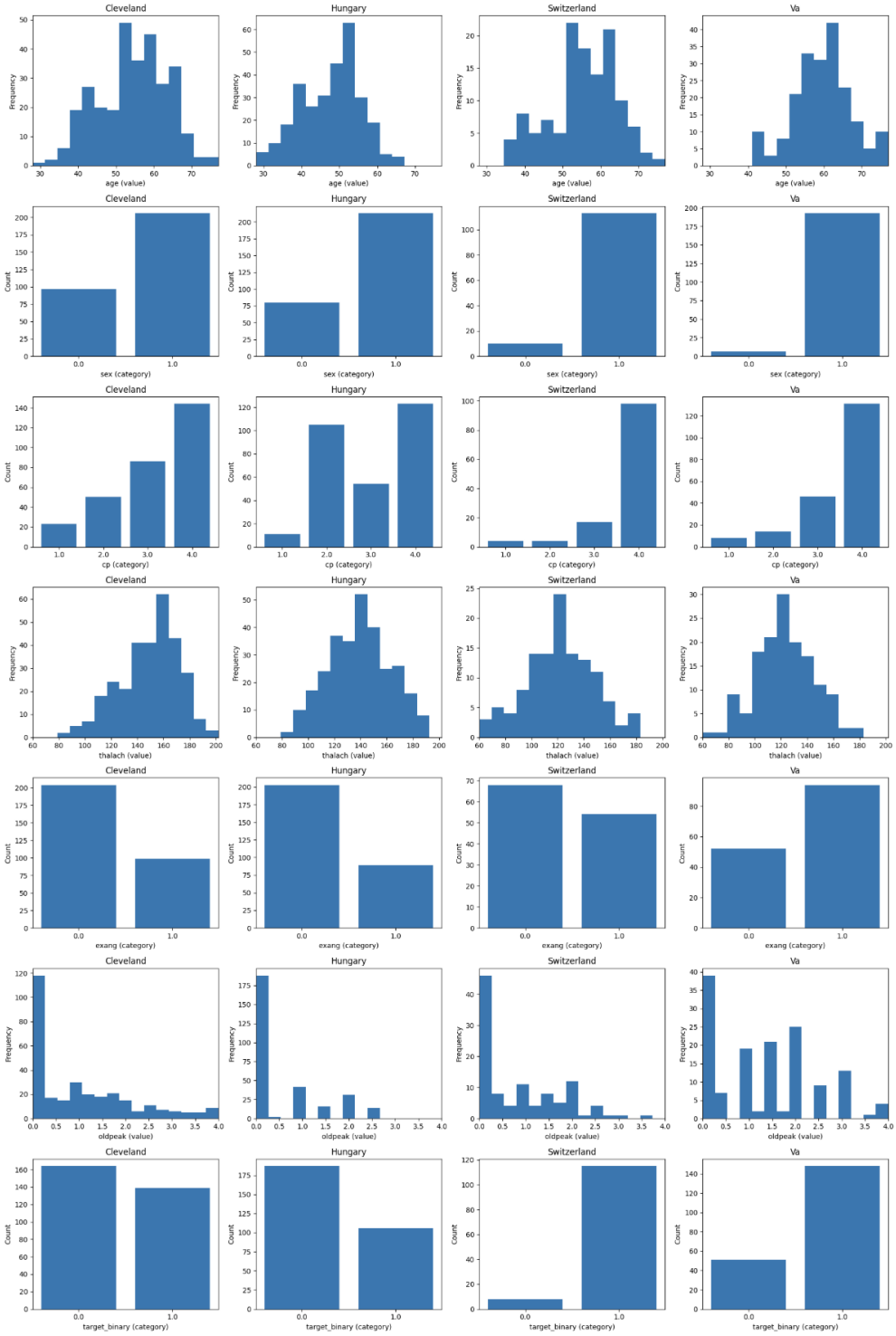


**Fig. 2** Feature distributions across cohorts showing variations in numerical and categorical variables.

Figure 2 shows that feature distributions differ across cohorts, particularly for age and thalach. These differences suggest the presence of distribution shift, which may affect model generalization.

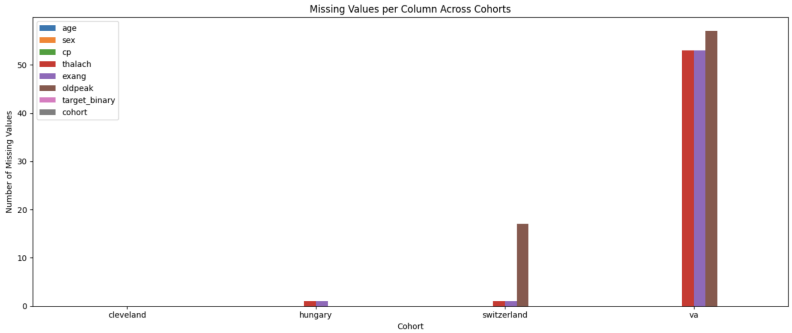


**Fig. 3** Missing values per feature across cohorts.

Figure 3 indicates that missing values are not evenly distributed across cohorts. The VA cohort contains the highest level of missingness, particularly in thalach, exang, and oldpeak, which may impact model performance and require careful preprocessing.

These variations further support the presence of distribution shift, which is expected to impact cross-cohort model performance.

A correlation heatmap was generated to examine relationships between features and the target variable.

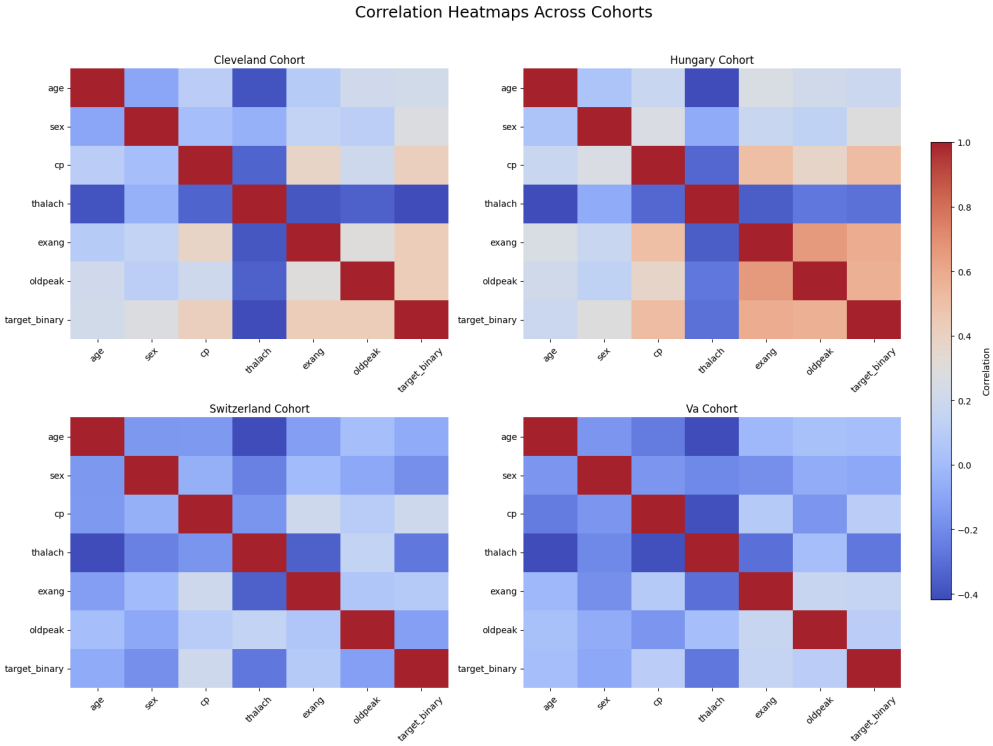


**Fig. 4** Correlation heatmaps for each cohort showing relationships between features.

Figure 4 shows that correlations vary across cohorts, indicating that feature relationships are not consistent across populations. This suggests that models may learn different patterns depending on the training data.

Boxplots were used to detect outliers in numerical features such as age, thalach, and oldpeak.

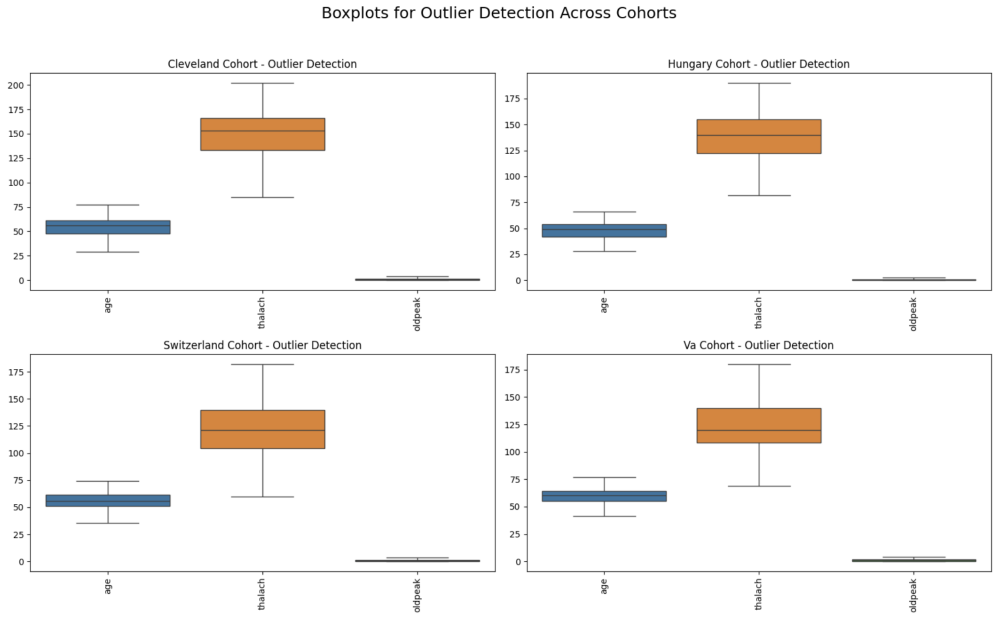


**Fig. 5** Boxplots for outlier detection across cohorts.

Figure 5 highlights the presence of outliers, particularly in thalach and oldpeak. These extreme values may introduce bias and affect model training and predictive stability.

Finally, scatter plots and grouped boxplots were used to examine the relationship between age, oldpeak, and the target variable.

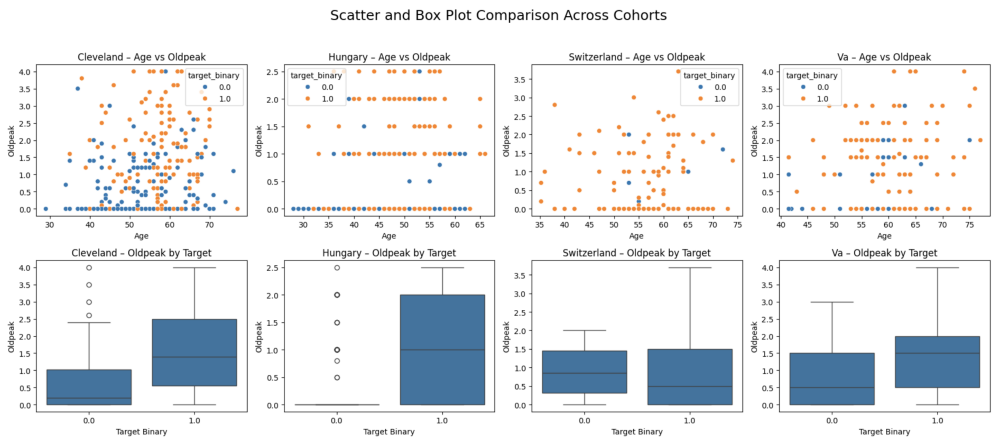


**Fig. 6** Scatter plots and boxplots showing the relationship between age, oldpeak, and the target variable.

Figure 6 shows that higher oldpeak values are generally associated with positive heart disease cases. However, the separation between classes is not consistent across cohorts, indicating variability in feature importance.

Overall, these visualizations provided important insights into the dataset. The analysis revealed the presence of class imbalance in the Switzerland and VA cohorts, which may influence evaluation metrics such as recall and F1-score. Differences in feature distributions across cohorts indicated the presence of distribution shift, suggesting that models trained on one cohort may not generalize well to others.

The missing value analysis showed that data completeness varies significantly across cohorts, with the VA cohort containing the highest level of missingness, which may affect model reliability. Correlation analysis demonstrated that relationships between features and the target variable are not consistent across cohorts, indicating that feature importance may differ depending on the population.

Outlier detection identified extreme values in features such as thalach and oldpeak, which may impact model training. Finally, the relationship analysis showed that higher oldpeak values are generally associated with positive heart disease cases, although this pattern varies across cohorts.

These findings highlight key challenges in the dataset, including class imbalance, missing data, and distribution differences, which help explain the differences in model performance observed during internal and external validation.

### 5.7 Feature Engineering and Model Design

In this study, no explicit feature reduction techniques such as Principal Component Analysis (PCA) were applied. This decision was made to preserve the interpretability of clinical features, which is important in medical prediction tasks.

However, preprocessing steps were applied to improve model performance. Numerical features were scaled using standardization, and categorical variables were encoded using one-hot encoding. These steps ensure that all features are in a suitable format for machine learning algorithms.

Logistic Regression includes regularization, which can help reduce overfitting and improve generalization. In this study, it was used as a baseline model due to its simplicity and interpretability.

No data augmentation techniques were applied, as the dataset consists of structured clinical data rather than images or text. Instead, model evaluation focused on cross-cohort validation to test robustness across different populations.

## 6 Results

### 6.1 Internal Validation Results

Internal validation was performed by training and testing the models within the same cohort using a 70/30 train–test split. The results showed that both Logistic Regression and Random Forest achieved consistently strong predictive performance within individual cohorts.

In general, Random Forest achieved slightly higher internal performance than Logistic Regression across several cohorts; however, this improvement was not consistent across all datasets. Despite its simplicity, Logistic Regression demonstrated competitive performance and served as a strong baseline model.

The Switzerland cohort shows unusually high performance, including perfect recall and lower ROC-AUC.

The internal validation results indicate that machine learning models can successfully learn patterns associated with heart disease within each dataset when the training and testing data come from the same population.

The detailed internal validation results for Logistic Regression and Random Forest are shown in Figures 7 and 8.

Internal Validation Results

| | model | cohort | n_total | internal_accuracy | internal_precision | internal_recall | internal_f1 | internal_roc_auc |
|---|---|---|---|---|---|---|---|---|
| 0 | LogisticRegression | cleveland | 303 | 0.824176 | 0.825000 | 0.785714 | 0.804878 | 0.890671 |
| 1 | LogisticRegression | hungary | 293 | 0.840909 | 0.800000 | 0.750000 | 0.774194 | 0.925781 |
| 2 | LogisticRegression | switzerland | 123 | 0.945946 | 0.945946 | 1.000000 | 0.972222 | 0.557143 |
| 3 | LogisticRegression | va | 199 | 0.800000 | 0.800000 | 0.977778 | 0.880000 | 0.765185 |

**Fig. 7** Internal validation results for Logistic Regression across cohorts.

Internal Validation Results

| | model | cohort | n_total | internal_accuracy | internal_precision | internal_recall | internal_f1 | internal_roc_auc |
|---|---|---|---|---|---|---|---|---|
| 0 | RandomForest | cleveland | 303 | 0.780220 | 0.761905 | 0.761905 | 0.761905 | 0.878523 |
| 1 | RandomForest | hungary | 293 | 0.829545 | 0.742857 | 0.812500 | 0.776119 | 0.915179 |
| 2 | RandomForest | switzerland | 123 | 0.945946 | 0.945946 | 1.000000 | 0.972222 | 0.514286 |
| 3 | RandomForest | va | 199 | 0.783333 | 0.796296 | 0.955556 | 0.868687 | 0.791852 |

**Fig. 8** Internal validation results for Random Forest across cohorts.

## 6.2 External Validation Results

External validation was performed by training the models on the merged Cleveland and Hungary datasets and evaluating them on the Switzerland and VA cohorts. This cross-cohort setup was used to assess how well the models generalize to unseen patient populations.

The results showed a clear decrease in performance compared to internal validation, showing higher performance when models are trained and tested on the same dataset. This performance reduction reflects differences in data characteristics between cohorts, where differences in patient characteristics and data collection procedures affect model reliability.

Logistic Regression maintained relatively stable performance across external cohorts, achieving accuracy between approximately 0.74 and 0.75, F1-scores between

0.83 and 0.84, and ROC-AUC values around 0.76. In contrast, Random Forest demonstrated lower external performance, with accuracy around 0.64 and F1-scores between approximately 0.73 and 0.77, indicating reduced generalization ability compared to Logistic Regression.

The detailed external validation results for Logistic Regression and Random Forest are shown in Figures 9 and 10.

External Validation Results

| | model | train_cohort | test_cohort | n_train | n_test | accuracy | precision | recall | f1 | roc_auc |
|---|---|---|---|---|---|---|---|---|---|---|
| 0 | LogisticRegression | cleveland+hungary | switzerland | 596 | 123 | 0.739837 | 0.966292 | 0.747826 | 0.843137 | 0.758696 |
| 1 | LogisticRegression | cleveland+hungary | va | 596 | 199 | 0.753769 | 0.861314 | 0.797297 | 0.828070 | 0.757750 |

**Fig. 9** External validation results for Logistic Regression on Switzerland and VA cohorts.

External Validation Results

| | model | train_cohort | test_cohort | n_train | n_test | accuracy | precision | recall | f1 | roc_auc |
|---|---|---|---|---|---|---|---|---|---|---|
| 0 | RandomForest | cleveland+hungary | switzerland | 596 | 123 | 0.642276 | 0.973333 | 0.634783 | 0.768421 | 0.765217 |
| 1 | RandomForest | cleveland+hungary | va | 596 | 199 | 0.648241 | 0.861111 | 0.628378 | 0.726562 | 0.725755 |

**Fig. 10** External validation results for Random Forest on Switzerland and VA cohorts.

The larger performance drop observed in Random Forest indicates reduced performance on external cohorts compared to Logistic Regression, making it less robust when applied to unseen populations.

## 6.3 Model Comparison

As shown in Figures 9 and 10, the differences in model performance between internal and external validation highlight the impact of distribution shift on generalization.

Overall, both Logistic Regression and Random Forest produced strong predictive results during internal validation. However, their performance differed under external validation. Random Forest achieved competitive internal performance but showed a larger decrease when evaluated on external cohorts.

In contrast, Logistic Regression demonstrated more stable performance across external cohorts, achieving higher accuracy and F1-scores on both Switzerland and VA datasets.

These results indicate that increased model complexity does not necessarily lead to improved generalization. While Random Forest is capable of modeling nonlinear relationships, it exhibited greater performance variability across cohorts compared to Logistic Regression.

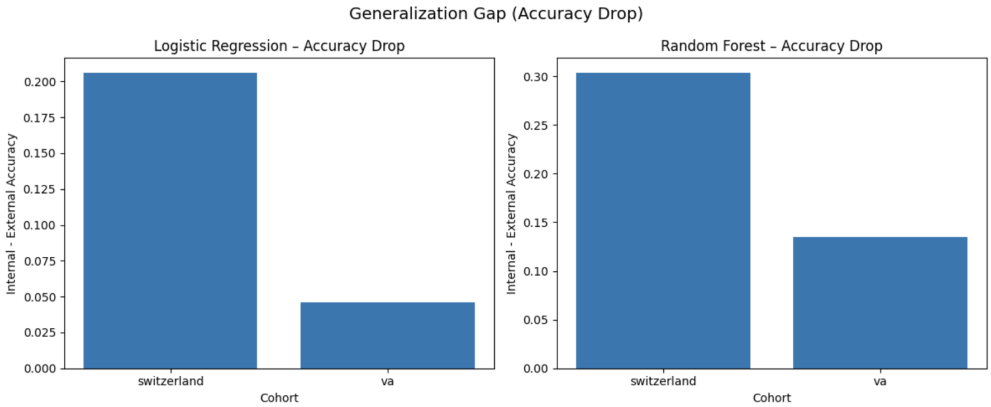


**Fig. 11** Generalization gap measured as the drop in accuracy from internal to external validation for Logistic Regression and Random Forest.

Figure 11 illustrates the reduction in accuracy when models are evaluated on external cohorts. Both models show a noticeable drop, with Random Forest experiencing a larger decline than Logistic Regression.

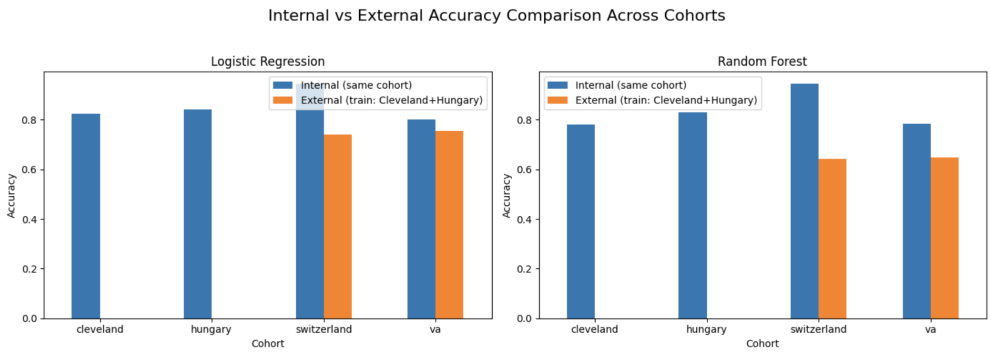


**Fig. 12** Comparison of internal and external accuracy across cohorts for Logistic Regression and Random Forest.

Figure 12 compares internal and external accuracy across cohorts. In all cases, external accuracy is lower than internal accuracy, confirming that models trained on one distribution do not fully generalize to others. Logistic Regression shows more consistent performance across cohorts, while Random Forest demonstrates greater variability.

These findings contradict Hypothesis 3, which assumed that ensemble-based models would be more robust across different populations. Instead, the results show that Random Forest did not generalize better than Logistic Regression under cross-cohort conditions.

Overall, these results highlight the importance of evaluating model performance beyond internal validation when selecting models for real-world clinical applications.

# 7 Discussion

The results demonstrate that distribution shift plays a significant role in reducing model performance across cohorts. Variations in feature distributions, class imbalance, and missing data patterns contribute to the observed generalization gap, highlighting the challenges of applying models trained on one population to another.

The comparison between F1-score and accuracy further reveals the impact of class imbalance. In some cohorts, higher F1-scores alongside lower accuracy suggest that models tend to favor positive predictions, indicating a bias toward the dominant class rather than true discriminative performance. This behavior reflects how imbalanced data can influence model evaluation and lead to misleading performance interpretations.

The results also show that Logistic Regression achieved better generalization than Random Forest across external cohorts. This suggests that simpler models may capture more stable and transferable patterns, while more complex models are more likely to learn cohort-specific relationships that do not generalize well to unseen populations.

This finding directly contradicts Hypothesis 3, which assumed that ensemble-based models would be more robust across different populations. Instead, the results indicate that increased model complexity does not necessarily improve generalization under distribution shift conditions and may, in some cases, reduce robustness.

These observations emphasize the importance of cross-cohort validation for evaluating model reliability. Without external validation, models may appear highly accurate but fail to perform consistently in real-world clinical settings.

Additionally, the results highlight that differences in cohort characteristics, including feature distributions and class balance, play a critical role in shaping model performance. This underscores the need to develop models that are robust not only to dataset size but also to underlying population variability.

From a practical perspective, these findings have important implications for clinical deployment. Models that fail to generalize across populations may introduce risks in decision-making, reinforcing the need for rigorous validation before real-world implementation.

Overall, the results suggest that model simplicity may improve robustness under distribution shift, while complex models remain more sensitive to variations in data distribution.

# 8 Limitations

This study has several limitations. The analysis was conducted using a single dataset, and although it includes multiple cohorts, it may not fully represent real-world clinical diversity. In addition, only two machine learning models were evaluated, and other algorithms may produce different generalization patterns. Future work could explore additional models and datasets to further validate these findings.

Future research could explore domain adaptation techniques, model recalibration strategies, and transfer learning approaches to improve cross-cohort generalization and robustness in clinical machine learning systems.

# 9 Conclusion

Previous studies show that machine learning models can achieve strong performance in predicting cardiovascular disease [5, 10]. However, models evaluated only through internal validation may perform worse when applied to new populations [6, 7]. Distribution shift has been identified as a key factor that can reduce the reliability of predictive models [11, 12]. For this reason, methodological guidelines emphasize the importance of external validation before deployment in clinical practice [1, 2].

In this study, machine learning models were developed using the UCI Heart Disease dataset and evaluated across four cohorts: Cleveland, Hungary, Switzerland, and VA Long Beach. Logistic Regression and Random Forest were trained and assessed using both internal validation and cross-cohort external validation. The results show that while both models perform well during internal validation, their performance declines when evaluated on different cohorts. Notably, Logistic Regression demonstrated more stable generalization than Random Forest, suggesting that simpler models may capture more transferable patterns across populations.

Overall, this study demonstrates that strong performance under internal validation does not guarantee reliable performance in real-world settings. The findings emphasize that external validation is essential for evaluating model robustness and ensuring safe deployment in clinical applications. Future work should focus on improving generalization through techniques such as domain adaptation, model recalibration, and training on more diverse datasets.

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